E214: ATLAS

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1 Introduction

For this laboratory course simulated Atlas data is analysed. Particles like electrons or muons are identified through the graphical program Atlantis. For quantitativ analysis the detector is calibrated with the Z-Boson mass and then used to measure the W-Boson mass.

1.1 *ATLAS detector*

The Atlas (A Toroidal LHC ApparatuS) detector is located at CERN (European Organization for Nuclear Research) to detect products of proton-proton collisions done through the LHC (Large Hadron Collider) at 14 TeV[1]. It consists of four main areas, the tracking system, the ECAL, the HCAL and the muon chambers. The tracking system is placed in the inner detector, being the closest to the collision with 3.3 cm from the beam line. There, four layers of silicon pixel detectors, one layer of silicon sensors and another layer of straws, made up of a gold plated tungsten wire surrounded by gas, get for a specific trajectory multiple hits through which the original trajectory can be reconstructed.[1] Because of an externally applied magnetic field, the trajectory is curved through the relativistic Lorentz force. This way the momentum of charged particles is measured. To measure the energy of the particles the ECAL and HCAL is introduced. The ECAL (Electromagnetic CALoremeter) is build with a material transmissive enough to let heavy particles like hadrons, muons or tauons pass, but so dense that electrons collide with the constituting atoms such that they emit bremsstrahlung. The photons produced by this, given a high enough energy, produce themself electron positron pairs. This forms a so-called electromagnetic shower, which deposits its charge in the ECAL. This charge is read out by anodes and corresponds to an energy in quantisations of the electron charge. Similarly this is done in the HCAL (Hadron CALorimeter), where the more heavy particles deposit their charge. Except, here no

EM showers are formed, but jets through the hadronization of fast quarks and gluons. Lastly in the muon chamber, the muons are detected.[2] Their trajectory also curves, due to a second external magnetic field, which is orthogonal to the field of the inner detector. When viewing the whole detector in polar coordinates, where the z-axis is directed along the particle beam, the outer magnetic field of the muon chambers is oriented along the rotation with the ϕ-angle and the inner magnetic field is oriented with the z-axis. A schematic illustration of the whole detector can be seen in Figure 1.

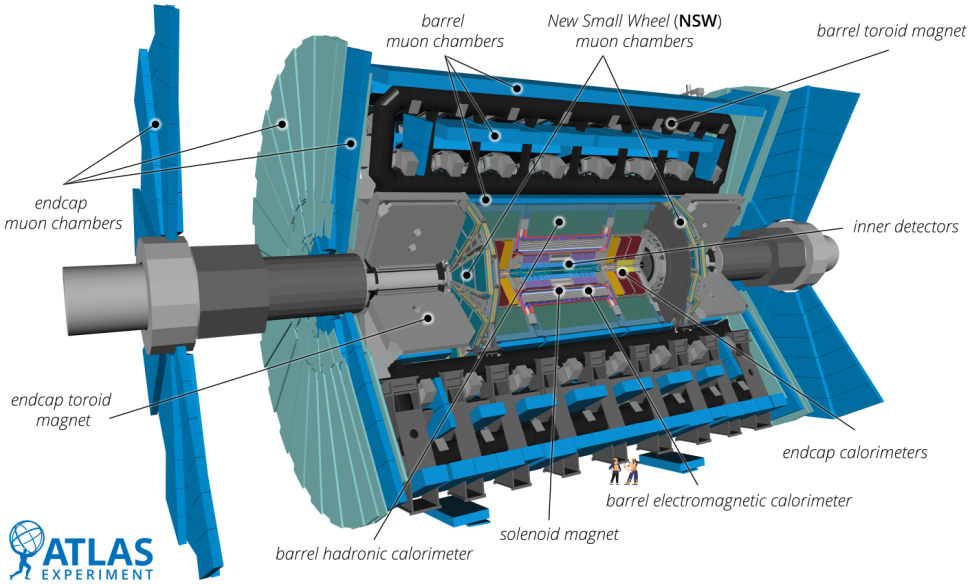

Figure 1: Schematic illustration of the ATLAS experiment, including new systems installed during Long Shutdown 2 (LS2).[1]

2 Event Displays and Calibration

The first task is to work with the graphical event display Atlantis, where different simulated events of an ATLAS detection are investigated. Furthermore the raw data needs to be calibrated with the well measured Z-Boson mass m_Z, though multiplicative scalar corrections of the energy for different angles θ and ϕ and other cuts. This is a good point to introduce the Lorentz invariant rapidity

$$y = \frac{1}{2} \ln \frac{E + p_z}{E - p_z}$$

Its usefulness gets apparent, when using the high energy approximation, where it gets reduced to the pseudo-rapidity

$$\eta = -\ln \tan \frac{\theta}{2}$$

Here it is seen that the non-Lorentz invariant angle θ can be replaced by the pseudo-rapidity, which will be Lorentz invariant for boosts along the z-axes.

2.1 *Questions to be answered before the lab*

2.1.1 *Decay of the Z^0 boson*

If a given Z^0 is at rest, the momentum of the electrons of the decay products $Z^0 \to e^+e^-$ are calculated through momentum and energy conservation. Using

$$0 = \vec{p}_{Z^0} = \vec{p}_{e^+} + \vec{p}_{e^-} \Rightarrow |\vec{p}_{e^+}| = |\vec{p}_{e^-}|, \tag{1}$$

$$m_{e^+} = m_{e^-} \tag{2}$$

$$\text{and } E^2 = p^2 + m^2 \tag{3}$$

the energy conservation

$$E_{Z^0} = E_{e^+} + E_{e^-}$$

can be rearranged to

$$|\vec{p}_{e^-}| = \sqrt{\frac{m_{e^+}^2}{4} - m_{e^-}^2}$$

Using $m_{Z^0} = 91.1880(20)\,\text{GeV}$ and $m(e^-) = 0.510\,998\,850\,69(16)\,\text{MeV}$, the momentum of the electron is

$$|\vec{p}_{e^-}| = 45.5940(10)\,\text{GeV}.$$

Here, the electron mass and uncertainty could have been neglected.

2.1.2 *Scattering reaction $e^+e^- \to \tau^+\tau^-$*

In the center-of-mass system of a $e^+e^- \to \tau^+\tau^-$ reaction, given $\sqrt{s} = E_{\text{COM}} = 5\,\text{GeV}$ and

$$E_{\text{COM}} = E_{\tau^-} + E_{\tau^+}, \tag{4}$$

Equations (1) to (3) can be used again to determine the tau momentum

$$|\vec{p}_\tau| = \sqrt{\left(\frac{\sqrt{s}}{2}\right)^2 - m_\tau^2} \tag{5}$$

For $m_\tau = 1776.86(12)\,\text{MeV}$ and no uncertainty on the center-of-mass energy, the momentum of the tauon is

$$|\vec{p}_\tau| = 1.756(68)\,\text{GeV} \tag{6}$$

p_e/GeV	E/GeV	$\frac{p_e}{E}$
-7.55	56.6	-0.133392226
-3.27	47.7	-0.0685534591
-89.35	86.5	-1.03294798
-50.35	0.6	-83.9166667
-28.62	28.1	-1.01850534
-32.92	64.3	-0.511975117
-34.3	0.6	-57.1666667
-36.4	33.8	-1.07692308
-55.46	48.8	-1.13647541
-69.17	49.2	-1.40589431

Table 1: Momentum, Energy and Ratio

2.2 *Graphic Display of Particle Reactions*

For this part the abilities of the graphical program Atlantis are investigated. Atlantis is a software designed for analysis of detected events from ATLAS. It allows to make cuts to filter out events that are not of current interest and identify particles by their traces in different parts of the detector. Therefor, it visualises the trajectories and deposed energy in the calorimeters.

2.2.1 *Identification of Common Particles*

Using example events for electrons, muons, tauons and photons the characteristic signatures in the detector are analyzed to gain an intuition on identifying those common particles.

2.2.2 *Energy-Momentum Relation of Electrons*

10 different Electrons from the electron-learning dataset are loaded and the in the inner detector detected momentum and in the ECAL deposed energy are determent and their ratio ploted. See Figure 2 and Table 1.

In the following when talking about the ratio, the absolute value of the ratio is meant, since the negative sign stems from the momentum, where the sign indicates the charge.

Two outliers are seen at ratios of -83.9 and -57.2. Those result from to low energies or to high momenta. Assuming the energy to be higher then the momentum, these events should not be possible and thus not correspond to electrons. Another explanation could be, that they are just faulty measurements or the electrons escaped the detector before hitting the ECAL.

Using only the measurements with a ratio close to 1, a vague gauss distribution can bee seen. Though for this conclusion more data points need to be taken. Also with the assumption, that the energy has to be higher as the momentum, only ratios below 1 should be expected. Observing the gauss distribution, the ratios being close to 1 and a few ratios being higher than 1, hints that the resolution of the detected energy and momentum is too low and the electrons move to close to the speed of light, to obtain only events where all measurements are correct.

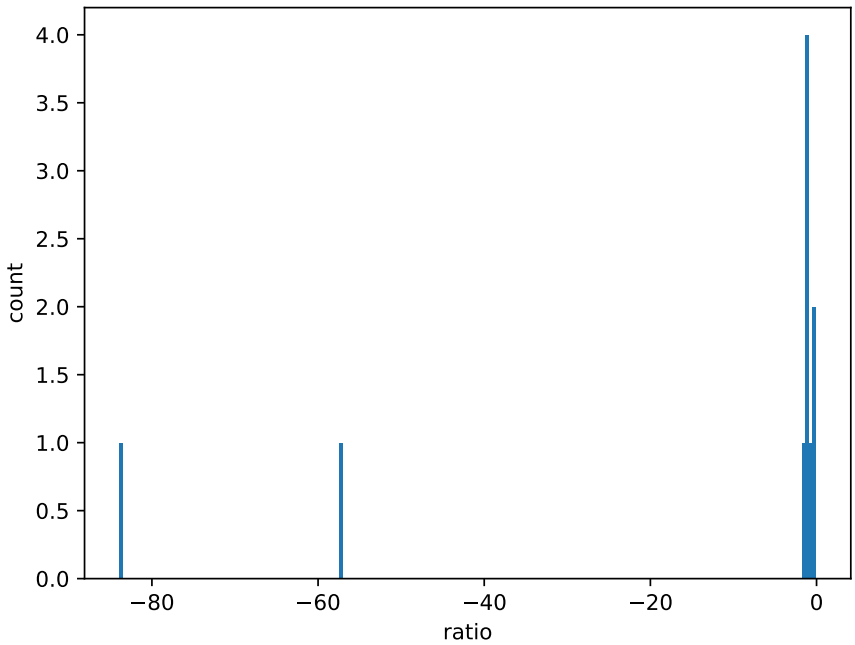

Figure 2: Momentum-Energy Ratio $\frac{|p|}{E}$

For assuming something like all the electrons being close to the speed of light or the determination of a resolution of the detector or distribution of the electron ratios, more data points are needed.

2.2.3 *Identification of a Mystery-Process*

A given set of events for a process, named `mystery-005-f.zip`, is provided which has to be identified. Firstly, the events are investigated regarding their missing transverse momentum. All events show a low E_T^{miss} in comparison to the total momentum $\sum E_T$; examples listed in Table 2. This excludes the process being an interaction with a W-Boson, since low missing transverse momenta suggests low amounts of neutrinos. Although jets are produced, the total deposed energy in the HCAL is relatively low with e.g. 8.6 GeV in event 11, 12.8 GeV in event 23 and 3.4 GeV in event 30. Thus no hadrons were produced and no gluon or tauon decayed. The primary observation of the data shows, that there is often a highly energetic muon-antimuon or electron-positron pair present. Three examples are listed in Table 2 and other incidences can be spotted in the Figures 14 to 18.

This can be seen, since a large number of events have high deposition of energy in the ECAL suggesting electrons, and trajectories in muon chambers suggesting multiple muons. Often, the most energetic electrons are opposite charged also suggesting them

Event	$\sum E_T$/GeV	E_T^{miss}/GeV	p_{p_1}/GeV	p_{p_2}/GeV	E_{j_1}/GeV	E_{j_2}/GeV
23	338.782	27.343	μ^+: 136.34	μ^-: -311.43	BL: 57.8	–
30	656.309	12.214	μ^+: 51.63	μ^-: -3506.59	TL: 212.0	–
119	205.393	11.980	e^+: 95.87	e^-: -56.98	T: 14.9	B: 1.0

Table 2: Events and their high-energy leptons with momenta p_p for particle $p \in \{e, \mu\}$ and jets j_i for jets $i \in \{1, 2\}$; B=Bottom, L=Left, T=Top. The named positions of the jets are seen in the lower panel of Figure 15, Figure 16 and Figure 17.

being electron-positron pairs. Further, when multiple muons present, the muons are also observed with opposite charge, suggesting muon-antimuon pairs.

Now having excluded gluon, tauon and W-Boson processes, while observing pair production of leptons, the Z^0 decay seems likely being the unknown process. The only, but large, contra argument is, that the hadronic branching ratio of the Z^0-Boson decay is 69.91(6) %[3]. So the majority of energy should be deposed in the HCAL, which is not observed here. A reason for the absence of energy in the HCAL, could be that the dataset has been previously filtered, where all hadronic events have been removed. Another interesting observation is the significant energy of the jets, see Table 2. A possible explanation could be, that the decayed Z^0 coupled beforehand to a hadron. Though, then a significant energy should be observed to be deposited in the HCAL. By now it should be noticed that there is no definite conclusion. The authors best guess is, that Z^0 coupled to a quark and then decayed into a electron-positron or muon-antimuon pair where all datasets have been removed where the jets deposed a significant amount of energy in the HCAL.

2.3 *Calibration of Electrons*

As for every detector, also ATLAS experiences some losses in the signal, namely the amount of energy deposed in the calorimeters in comparison to the true energy of the measured event or particle. It is assumed that those losses are caused by irregularities of the used material and also the fact that particles take longer to cross the detector, when they pass in a steeper angle tangent to the detector surface of a given layer. Thus already by the latter, it can be assumed that the losses are especially depended on the pseudo-rapidity η and also, less significant, on the angle ϕ and some other cuts. Further it can be assumed that the loss is dependent on the momentum.

To correct the detected raw energy, it is multiplied by a certain coefficient. For the calibration, the $Z^0 \to e^- e^+$ decay is used, where the detected energy of the electrons has to be the mass of the Z^0-boson $m_{Z^0} = 91.1876(21)$ GeV[3]. Therefore the calibration is complete when those match. The uncalibrated energy spectrum is shown in Figure 3.

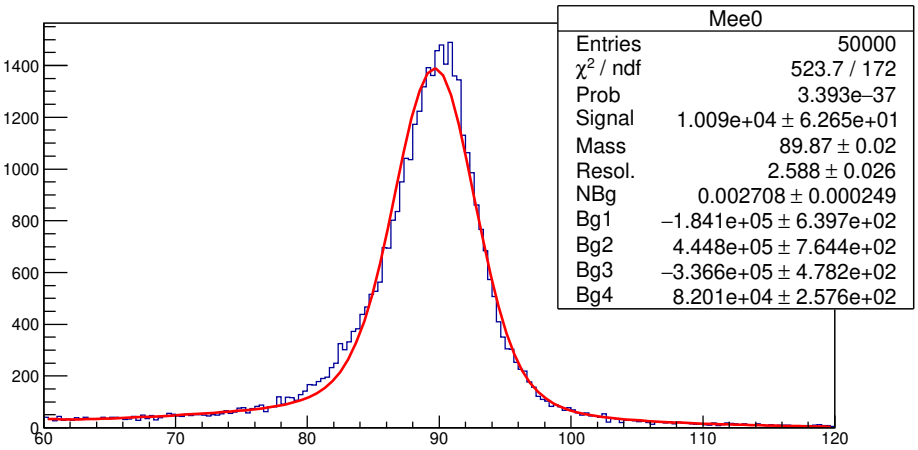

Figure 3: Energy spectrum with uncalibrated energies and no cuts.

In addition, the distribution needs to follow the Breit-Wigner function

$$F(M_{ee}) = \frac{\Gamma_Z^2/4}{(M_{ee} - M_Z)^2 + \Gamma_Z^2/4}, \tag{7}$$

which is an approximation of the more general function of eq. 6.1 in the lab manuel[2]. To account for smearing, the function is convoluted with the gauss distribution. This yields the Voigt function. Thus the calibration is complete when the energy of the electrons match the boson mass, i.e. the peak of the Voigt function is at the boson mass, and the Voigt function function yields a high resolution.

To achieve this, the function is repeatedly only fitted to a cut representing the correction. For example, when determining the coefficient for an if-statement, i.e. cut, `else if (fabs(eta)>2.0 && fabs(eta)<2.5) energy = energy * 91.2/89.9;`, the fit is done without this correction, but with a cut corresponding to `z.Fit("2.0TMath::Abs(el_eta) && TMath::Abs(el_eta)<2.5")`. Here the fit yields the mass 89.9 GeV, thus it has been put into the statement as a ratio with the true mass, to yield the coefficient to correct the raw mass to the true mass. The energy spectra with different cuts and uncalibrated energies are shown in Figure 4 as an example.

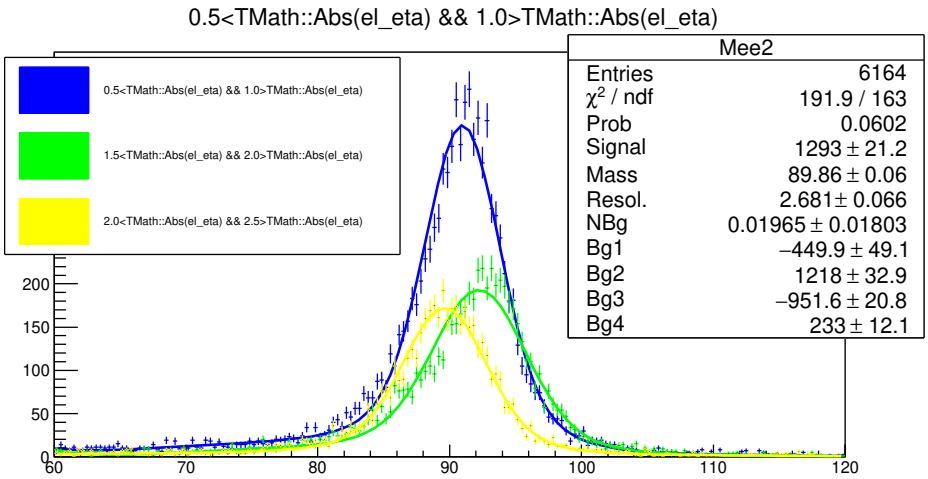

Figure 4: Energy spectrum with uncalibrated energies for different cuts.

Using this procedure the following calibration-code is obtained:

```
#include "math.h"
#include "TMath.h"

double ElecCalib(double e_raw, double pt, double eta,
   double phi, double etiso, double eoverp, double mindrjet)
{
  double dummy=pt*eta*phi*etiso*eoverp*mindrjet;
  double energy = e_raw;


  if (fabs(eta)>0.0 && fabs(eta)<0.5) energy = energy * 91.2/91.5;
  else if (fabs(eta)>0.5 && fabs(eta)<1.0) energy = energy * 91.2/91.2;
  else if (fabs(eta)>1.0 && fabs(eta)<1.5) energy = energy * 91.2/90.9;
  else if (fabs(eta)>1.5 && fabs(eta)<2.0) energy = energy * 91.2/92.5;
  else if (fabs(eta)>2.0 && fabs(eta)<2.5) energy = energy * 91.2/89.9;

  if (phi>-0.5*3.2 && phi<-0.25*3.2) energy = energy * 91.2/89.9;
  else if (phi>-0.25*3.2 && phi<-0.*3.2) energy = energy * 91.2/89.8;
  else if (phi>0.0*3.2 && phi<0.25*3.2) energy = energy * 91.2/89.8;
  else if (phi>0.25*3.2 && phi<0.5*3.2) energy = energy * 91.2/89.9;

  else if (phi>0.5*3.2 && phi<0.75*3.2) energy = energy * 91.2/89.9;
  else if (phi>0.75*3.2 && phi<1*3.2) energy = energy * 91.2/89.8;
  else if (phi>-1*3.2 && phi<-0.75*3.2) energy = energy * 91.2/89.9;
  else if (phi>-0.75*3.2 && phi<-0.5*3.2) energy = energy * 91.2/89.8;
```

```
  if (pt>0 && pt<25) energy = energy * 91.2/89.8;
  else if (pt>25 && pt<50) energy = energy * 91.2/91.1;
  else if (pt>50 && pt<75) energy = energy * 91.2/92.1;
  else if (pt>75 && pt<100) energy = energy * 91.2/91.8;
  else if (pt>100) energy = energy * 91.2/92.1;

  return energy;
}
```

Usually an open bin has to exist for every cut, but ϕ is only defined for $[-\pi, \pi]$ and $|\eta|$ has no data beyond 2.5.

As result of the calibration Figure 5 is obtained.

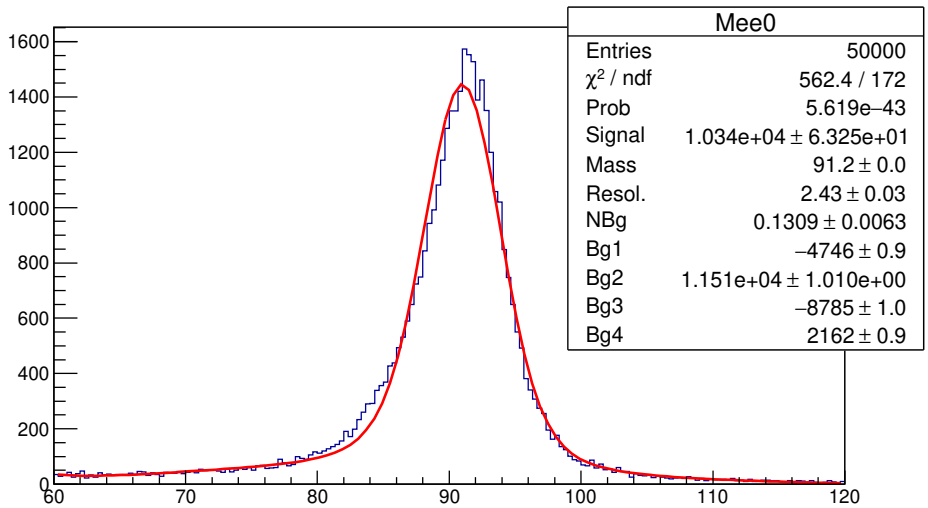

Figure 5: Calibrated energy spectrum.

Here a mass of 91.2 GeV and resolution of 2.43(3) GeV is obtained. In comparison to the uncalibrated spectrum, only a small improvement of the resolution 2.588(26) GeV is observed. Though, per definition of the used calibration method, the raw, i.e. original, boson mass 89.897(20) GeV is massively improved to the exact value.

3 Theory for Day 2

W-bosons predominantly decay into quark-antiquark pairs, but unfortunately this decay mode is not distinguishable from the general QCD background, and the measurement of jet directions and energies is not exact. Hence, we will instead use the decay of the W-boson into lepton and lepton-neutrino pairs. In these $W \to e\nu_e$ and $W \to \mu\nu_\mu$ decays, the leptons can be easily detected in ATLAS, however this is not

true for the neutrinos whose presence and properties must be observed indirectly by the imbalance of the transverse momentum.

When the W-boson decay occurs, in its rest frame, the angular distribution of the leptons with reference to the axis of the incoming partons is given by

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\cos\theta} = 1 + \cos^2\theta$$

By doing a transformation of variables, we can obtain a distribution of the transverse momentum:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}p_\mathrm{T}} = \frac{\mathrm{d}\sigma}{\mathrm{d}\cos\theta}\left|\frac{\mathrm{d}p_\mathrm{T}}{\mathrm{d}\cos\theta}\right|^{-1} = \frac{\mathrm{d}\sigma}{\mathrm{d}\cos\theta}\frac{2p_\mathrm{T}}{m_W}\frac{1}{\sqrt{\frac{1}{4}m_W^2 - p_\mathrm{T}^2}}$$

In the lab frame, there will simply be a Lorentz boost along the beam axis and hence the transverse components will remain unchanged. So, what we would observe from the above expression is the existence of a pole at $p_\mathrm{T} = m_W/2$, which is called a Jacobi peak. Measuring the location of the Jacobi peak thus lets us calculate the mass of the W-boson.

Ideally, the peak would be at a single point, however in practice it is smeared due to the finite resolution of the detector, the W-decay width, and the transverse momentum p_T^W of the W-boson. This can be dealt with to some extent by measuring instead the location of the half-maximum of the peak. Since the smearing effects affect the distribution symmetrically, the position ofthe half-maximum will remain unchanged.

4 Questions to be answered before the lab of Day 2

4.1 *Estimated Transverse Momentum*

The 'W to e nu' dataset contains the properties **ptmisx** and **ptmisy**, which are the x and y components of the $E_\mathrm{T}^\mathrm{miss}$ vector respectively. We can use them to calculate the length of the vector itself,

$$E_\mathrm{T}^\mathrm{miss} = \sqrt{(\mathbf{ptmisx})^2 + (\mathbf{ptmisy})^2}$$

which is the missing transverse energy (also obtainable from the property **etmis**). Now, in the relativistic limit, the energy and absolute value of the momentum are the same, hence we can treat this missing transverse energy as the absolute momentum of the neutrino emitted by the W-boson's decay.

This means that the transverse momentum of the W-boson (equivalent to the property **ptw**) is

$$\mathbf{ptw} = -E_\mathrm{T}^\mathrm{miss} = -\sqrt{(\mathbf{ptmisx})^2 + (\mathbf{ptmisy})^2}$$

4.2 *Propagation of Correlated Errors*

The simplest form of Gauss's error propagation law is

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2} \tag{8}$$

where $f(x, y)$ is a function of the (here uncorrelated) variables x and y, with σ_x^2, σ_y^2 being their variances. When x and y are correlated however, we introduce another term under the square root,

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2\sigma_{xy} \left(\frac{\partial f}{\partial x}\right)\left(\frac{\partial f}{\partial y}\right)}$$

where $\sigma_x y$ is the off-diagonal element of the covariance matrix. If however, the value of this is not known, we can find an upper estimate of the uncertainty of f by simply adding up the uncertainties of x and y.

5 Day 2

5.1 *Setup*

We set up the second part of the experiment by loading up the `w` object into the ROOT session at the beginning, running the following 2 lines of code: `.L Wenu.C+` \\ `Wenu w`.

5.2 *Verification of electron calibration*

Next, to verify that the electron calibration from the previous part was applied correctly, we ran the `FitZee()` command (working similar to the fit function of the previous part) to measure the $Z^0 \to ee$ mass. Fitting the whole dataset using the `w.FitZee("")` gave us the plot in Figure 6. This fit gave a weight of the Z-boson as 91.16 ± 0.01 GeV. This is close to the true value of 91.1876 ± 0.0021 GeV, however the uncertainty in our plot is an order of magnitude larger.

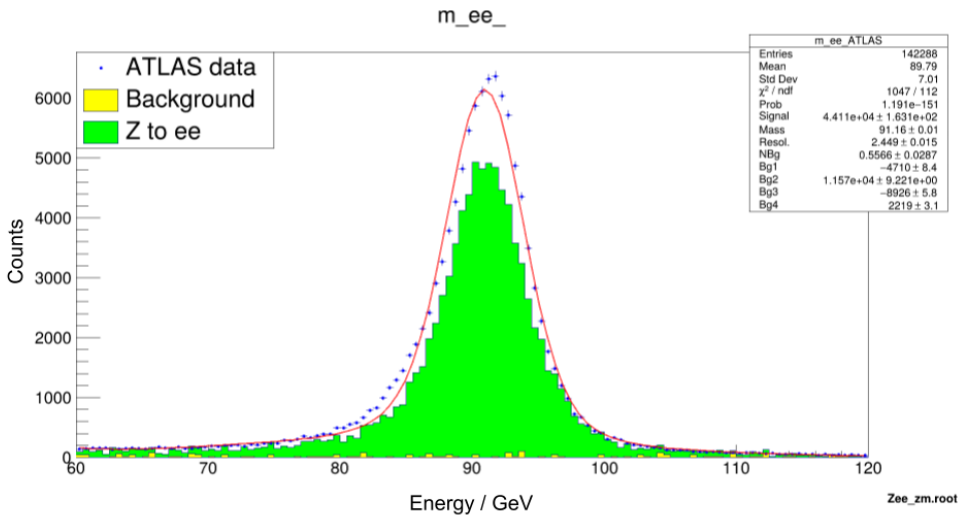

Figure 6: Fit of the Z^0 peak of ATLAS data using the electron calibration from the previous part; the mass is determined to be 91.16 ± 0.01 GeV

5.3 *Inspection of Kinematic Variables*

With the electron calibration checked, we began plotting the various observables of the 'W to e nu' dataset. Using the command (with general syntax)

```
w.PlotW("variable", bin-number, lower boundary, upper boundary, "cut selection");
```

we obtained a stack plot of the ATLAS data alongside a simulated result. We observed in these plots a mismatch between the simulated and the actual data, as can be seen in Fig. 7 for example, caused due to the QCD background not having been properly normalized yet. To correct for this, we attempted by eye to match the simulated data curve with the curve from the actual 'W to e nu' data. This correction was implemented by the command

```
w.SetQCDScaleFactor(VALUE));
```

where `VALUE` was determined by us to be the factor $1.85/3.2 \times 192/225 \approx 0.493$. The corrected graph can be seen in Fig. 8.

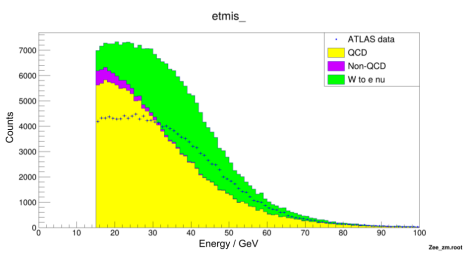

Figure 7: `etmis` (Missing Transverse Energy) data from ATLAS stack plot along with blue dotted curve showing simulated data; the QCD background has not been normalized (i.e. set to a preliminary normalization factor of 1).

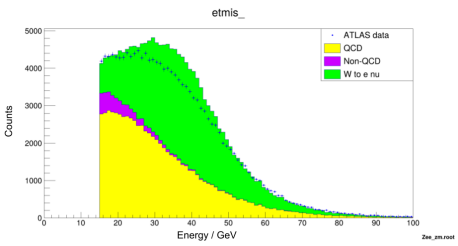

Figure 8: `etmis` (Missing Transverse Energy) data from ATLAS stack plot along with blue dotted curve showing simulated data; the QCD background has been normalized.

With the QCD normalization accounted for, we began looking at the plots of the different variables to identify those sections of it with a low QCD background that we could use for out cut selections later in the experiment. For example, plotting the number of jets (using the property `njets`), showed us the plot in Fig. 9, where the high amounts of QCD background when looking at any number of jets greater than zero informed us that `njets == 0` was a useful cut we could make. This could be further verified by looking at the plots of other variables when this cut was applied, and noting the high or low amounts of QCD background depending on the cut chosen (see Fig. 10 and Fig. 11).

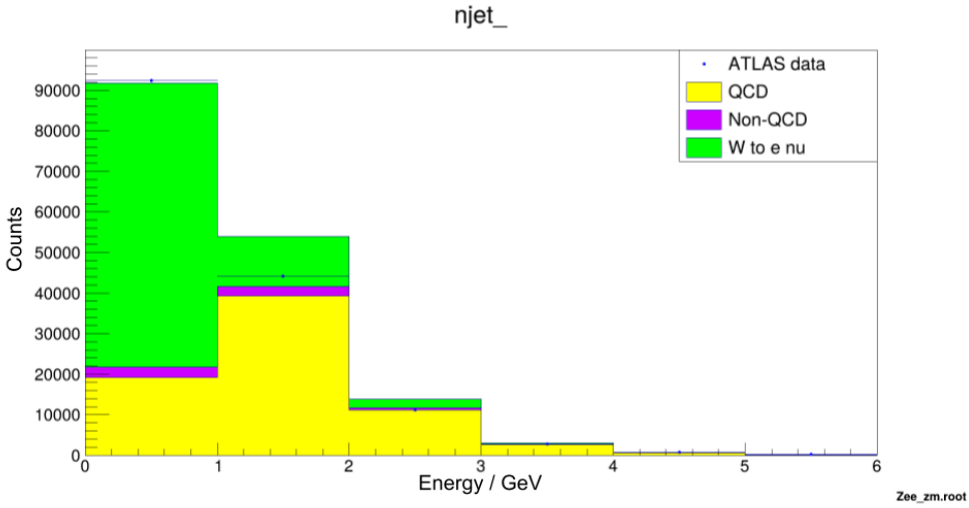

Figure 9: `njet` (Number of Jets) data from ATLAS stack plot along with blue dotted curve showing simulated data (after QCD normalization); the data for more than 0 jets shows a significant QCD background-to-signal ratio

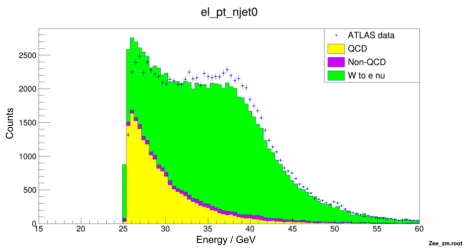

Figure 10: `el_pt` (Transverse Momentum of Electron) data from ATLAS stack plot along with blue dotted curve showing simulated data (after QCD normalization); a cut to only show that data where `njet==0` has been applied

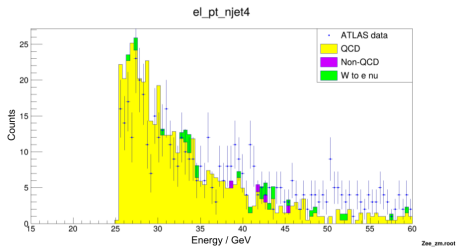

Figure 11: `el_pt` (Transverse Momentum of Electron) data from ATLAS stack plot along with blue dotted curve showing simulated data (after QCD normalization); a cut to only show that data where `njet==4` has been applied, note the greater presence of QCD background

Similarly, we attempted to find similar regions in the plots of the other variables which showed a low QCD background which we could use for our cuts, and which showed a prominent Jacobi peak in the `el_pt` plot. We finally settled on the following list of cuts: `njet==0, etmis>35, el_etiso>1, el_energy>40`. To apply this on all future fittings of the data, we ran the command

```
w.SetCutSelection("cut selection");
```

with `"cut selection"` replaced by the cut parameters we had chosen.

5.4 *Gauge curve W-boson mass versus half-maximum point*

Finally, we applied these cuts on to the different 'W to e nu' simulated data sets using the following set of commands (run as a ROOT macro):

```
TCanvas* c1 = new TCanvas("c1");
c1->Clear();
c1->Divide(4,2);
w.SetQCDScaleFactor(1.85/3.2 × 192/225));
w.SetCutSelection("njet==0, etmis>35, el_etiso>1, el_energy>40");
c1->cd(1);w.GetHalfMaximum("MCW75",xmin,xmax);
c1->cd(2);w.GetHalfMaximum("MCW78",xmin,xmax);
c1->cd(3);w.GetHalfMaximum("MCW79",xmin,xmax);
c1->cd(4);w.GetHalfMaximum("MCW80",xmin,xmax);
c1->cd(5);w.GetHalfMaximum("MCW81",xmin,xmax);
c1->cd(6);w.GetHalfMaximum("MCW82",xmin,xmax);
c1->cd(7);w.GetHalfMaximum("MCW85",xmin,xmax);
```

These commands create a 4-by-2 subplot of images which are subsequently filled in by the 7 different plots from the 7 different simulated datasets (i.e. the first subplot is of the fit applied to the simulated data with mW = 75 GeV, the second subplot is of the fit applied to the data with mW = 78 GeV, and so on). The QCD Scale Factor setting and cut selections are applied directly in the macro to ensure that any fitting that happens has those set to the values and factors we had previously decided, instead of being affected by any change that may have occurred in the outside ROOT environment.

In each simulated `el_pt` plot, a half maximum fit was applied on the data with x-coordinates between `xmin` and `xmax`, which we determined by eye to be 30 and 55 respectively for most appropriate fittings, for all the datasets. (The plots are shown in Fig. 13 as the first seven subplots. The last two subplots are those of the fits done on the actual 'Wenu' and 'Zee' datasets, explained ahead. These plots were generated by slightly modifying the above macro to be a 3-by-3 plot instead of a 4-by-2 one.) We thus obtained the value of the half-maximum point for each value of mW given in the datasets. This let us construct a gauge curve relating the two (Fig. 12).

mW / GeV	Half-maximum / GeV
75	40.97
78	42.55
79	43.01
80	43.51
81	44.09
82	44.43
85	45.90

Table 3: W-boson masses from the different data sets, and the corresponding half-maxima

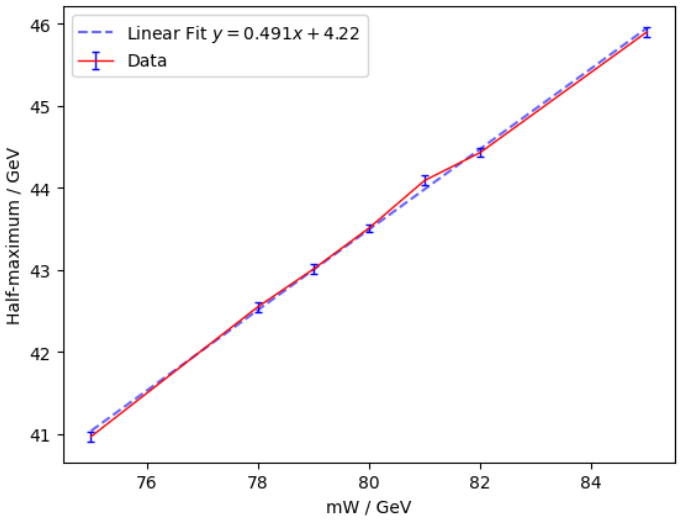

Figure 12: Above tabular data plotted with error bars, along with a best fit line obtained from `scipy.curvefit()`

We now applied the fit onto the true dataset `Wenu`, using the same fit, QCD, and cut parameters, and obtained a half-maximum here of 44.23 GeV (seen as the eighth subplot in Fig. 13). Plugging this into the expression for the gauge curve we fit above gave us a base value for the mass of the W-boson of **81.50 GeV**.
Before looking closer at this obtained value to determine the uncertainty in it, we also did a quick check by plotting the fit with the parameters used so far on the 'Z to ee' plot, to observe if we obtain the expected Z-boson mass here (the plot can be seen as the ninth subplot in Fig. 13). We got here a half-maximum of 49.74 GeV, which corresponds via the gauge curve to a Z-boson mass of 92.73 GeV, which is close enough to the true mass to show that the parameters we had chosen were quite good.

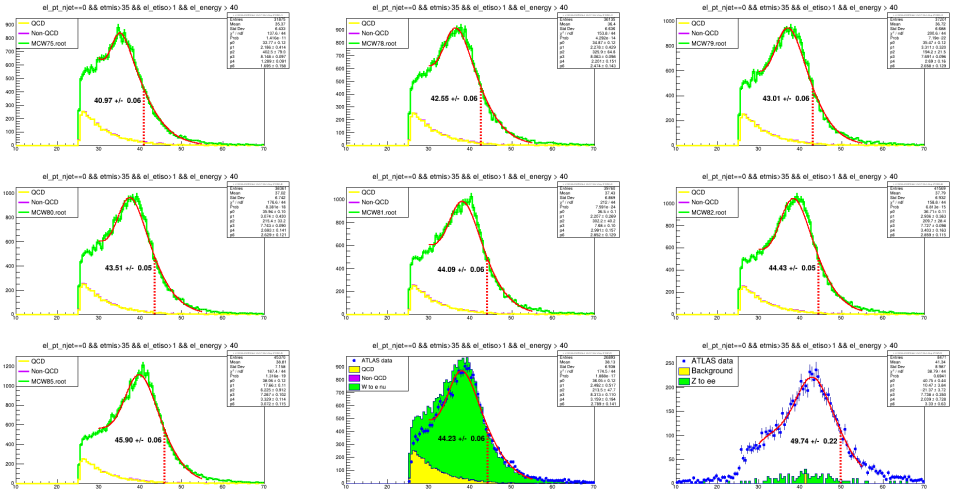

Figure 13: Plots showing the different datasets with the half-maximum fit, each showing the corresponding value. The first seven subplots are for the simulated datasets where the W-boson had a range of masses (listed in the legend of each subplot). The eight subplot is for the actual dataset where the W-boson had its true mass. The ninth subplot is for the Z-boson dataset.

5.5 *Other systematic effects and error sources*

To find the statistical uncertainty in our value of the mass of the W-boson, we used the fit formula obtained above (half-max = $0.491 m_W + 4.22$ or rather, this equation flipped to have m_W on one side) and the simple form of Gauss's error propagation law Eq. 8 (i.e. assuming no correlation). This gives us a statistical uncertainty of 1.78 GeV.

For the systematic uncertainty, we approached this by varying the various parameters we had set, first setting them to those values that most loosely fit the data while still being reasonable, then setting them to those values that most tightly fit the data. We changed the cuts we had made, the value of the QCD normalization, and the range of x-coordinates on which the fit was applied. Each set of parameters was changed keeping the other two sets constant. (For the QCD systematic error, when calculating the scale factor by eye originally, we attempted to match the blue dotted curve with the green stack plot as best as we could with reference to the ticks along the Y-axis (see Fig. 7). When we had settled on a scale factor, we set the least count of the Y-axis at that point as the error, which meant our estimate could've been about 4% greater or less, which is now reflected in this multiplication by 24 (or 26)/25.) The values of m_W that resulted from this are detailed below in Table 4:
As can be seen, the variation in the QCD scale factor changes the measured W-boson mass very little, with both these values and our original value lying comfortably within their uncertainties. Changing the cuts and the range of x-coordinates changes the values of m_W a little more, but again, they are close enough to our original value

Parameters changed	Change made	m_W / GeV
Cuts	`etmis>25, el_etiso>0.4, el_energy>20` (Loose)	80.89 ± 1.56
	`etmis>50, el_etiso>1.4, el_energy>60` (Tight)	82.93 ± 4.46
QCD Scale Factor	$(1.85/3.2 \times 192/225) \times 24/25$ (Loose)	81.49 ± 1.78
	$(1.85/3.2 \times 192/225) \times 26/25$ (Tight)	81.52 ± 1.78
Range of x-coordinates	25 to 70 (Loose)	83.29 ± 2.05
	33 to 50 (Tight)	81.10 ± 2.95

Table 4: The variation of the various parameters to a loose and tight extent, along with the corresponding resulting calculations for the mass of the W-boson.

to not fall outside the uncertainties.

Finally, comparing the value we obtained to the expected value of 80.403 ± 0.029 GeV, we see that this value lies within the statistical uncertainties we calculated. So, our measured value of $m_W = 81.50 \pm 1.78$ GeV is good reproduction of the true, expected value.

6 Conclusion

In this experiment, we studied data from the ATLAS detector at CERN, learning how to identify the tracks of different elementary particles in the detector. With this, we were able to identify the mystery process given to us as most likely to be a Z^0 decay based on the jets and lepton-anti-lepton pairs observed; this identification however is not certain since the more likely decay of a Z boson to hadrons is not observed in the expected quantities.

We then calibrated the electron data to match the observed mass of the Z-boson with the true, expected mass. The calibration was successfully done, resulting in an improvement of the mass from 89.897(20) GeV to 91.2 GeV.

Finally, we calculated the mass of the W-boson by measuring the location of the Jacobi peak in the distribution of the transverse momentum of the electron. This resulted in us measuring a value of 81.50 ± 1.78 GeV, within the uncertainties of which the true value of 80.403 ± 0.029 GeV falls.

7 Appendix A

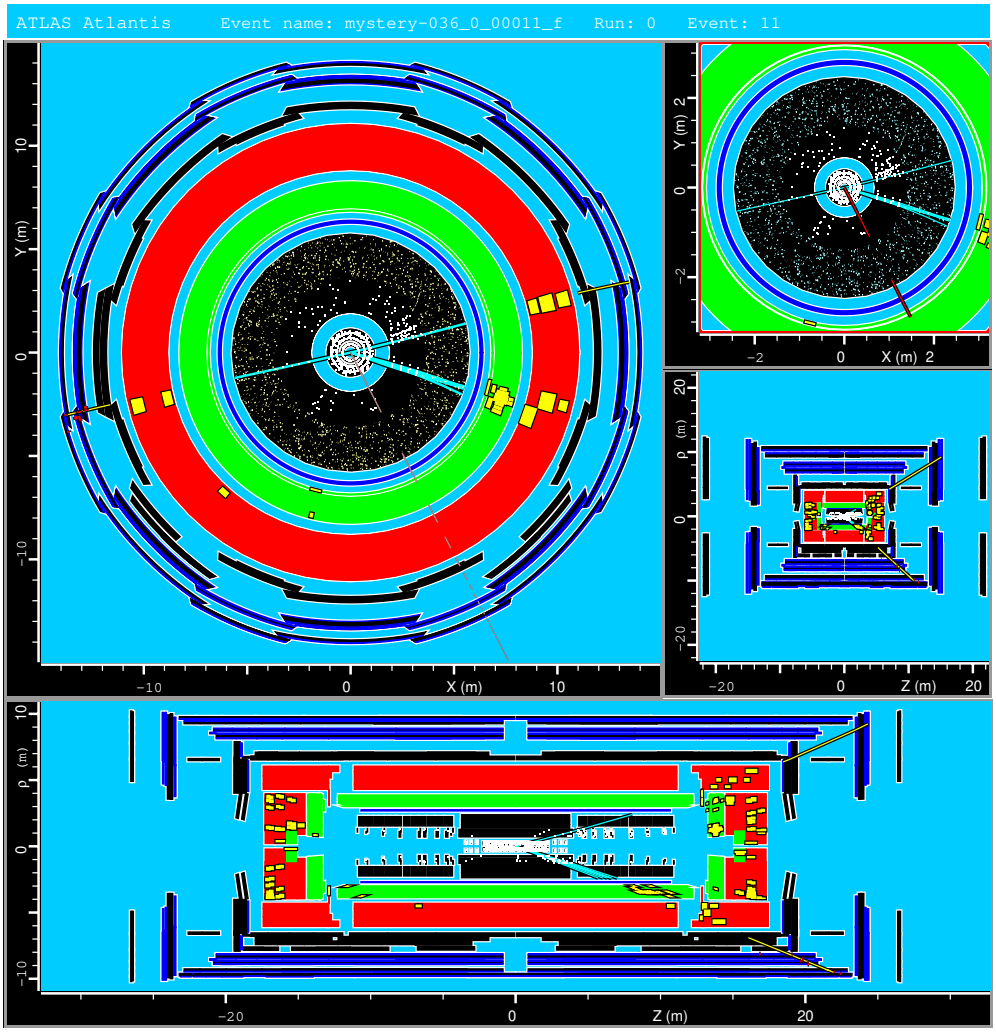

Figure 14: Event 11 with two detected muons

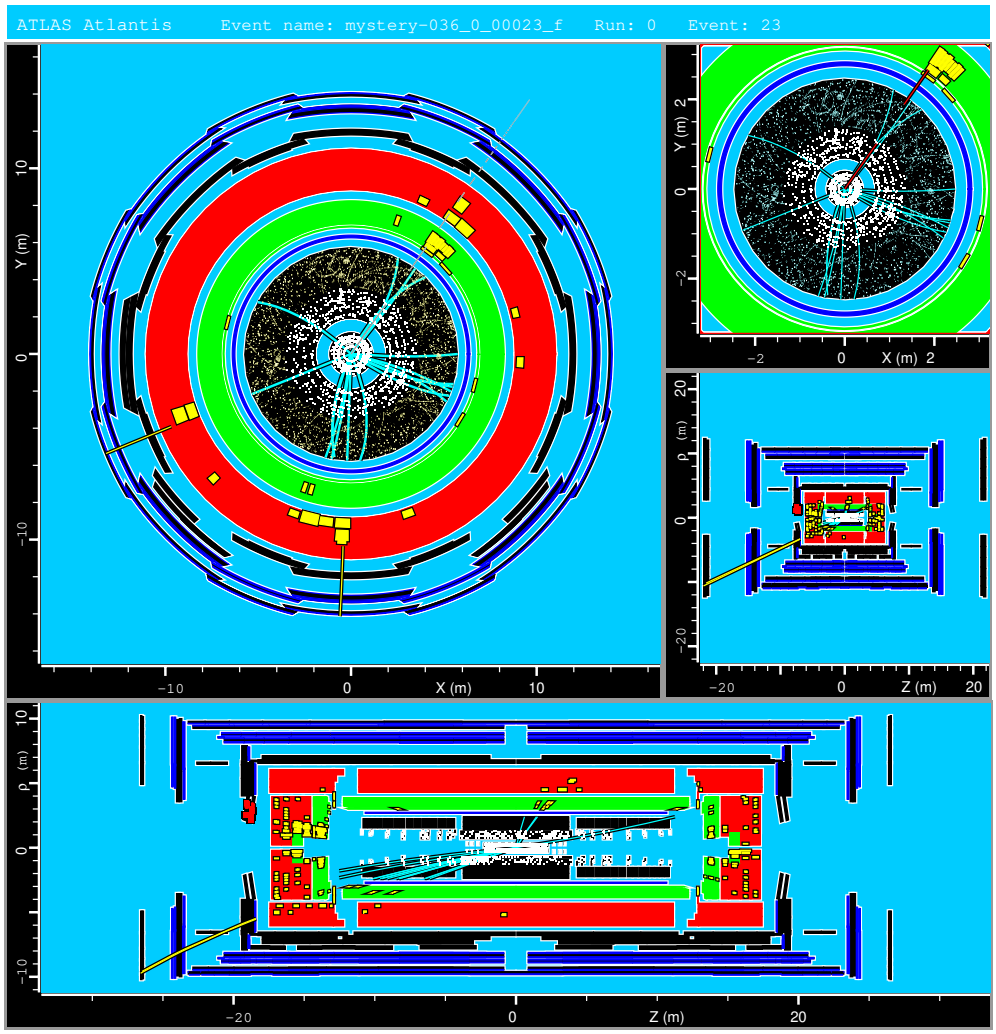

Figure 15: Event 23 with two detected muons an jets

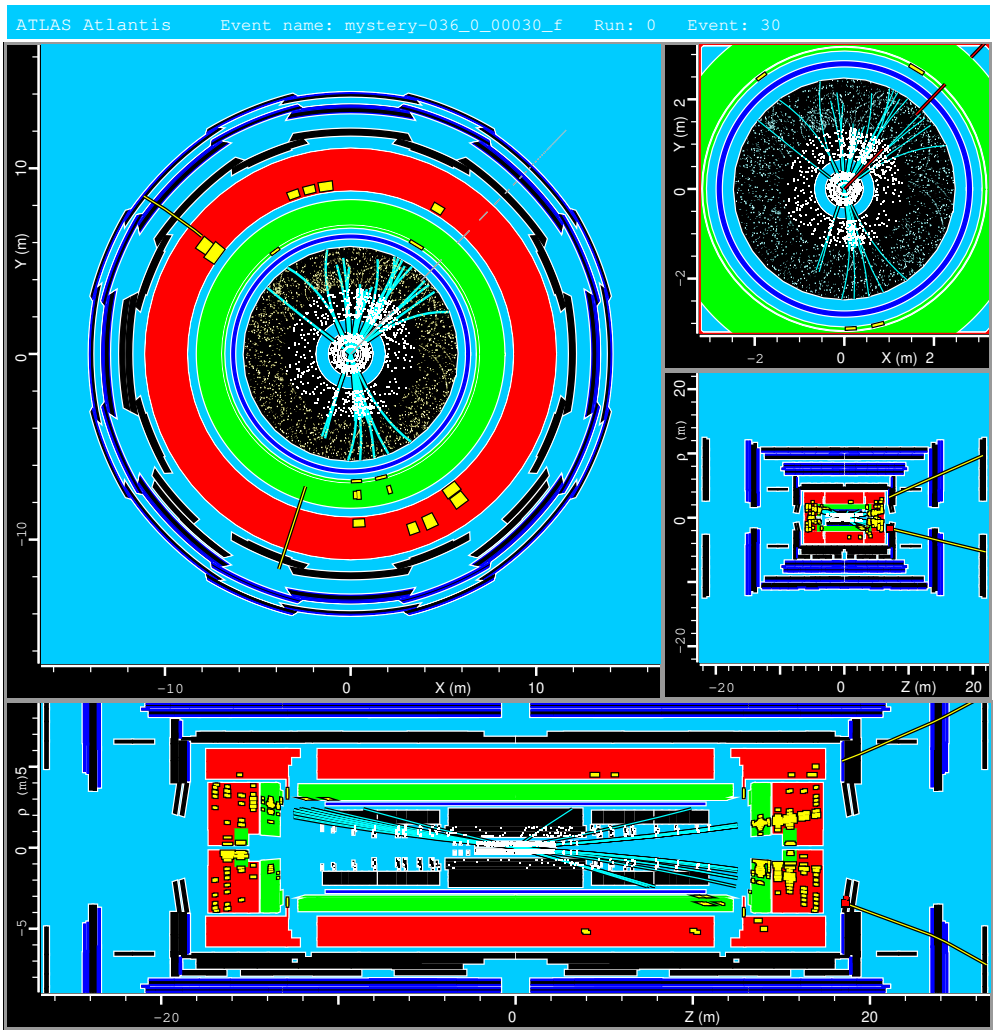

Figure 16: Event 30 with two detected muons and jets

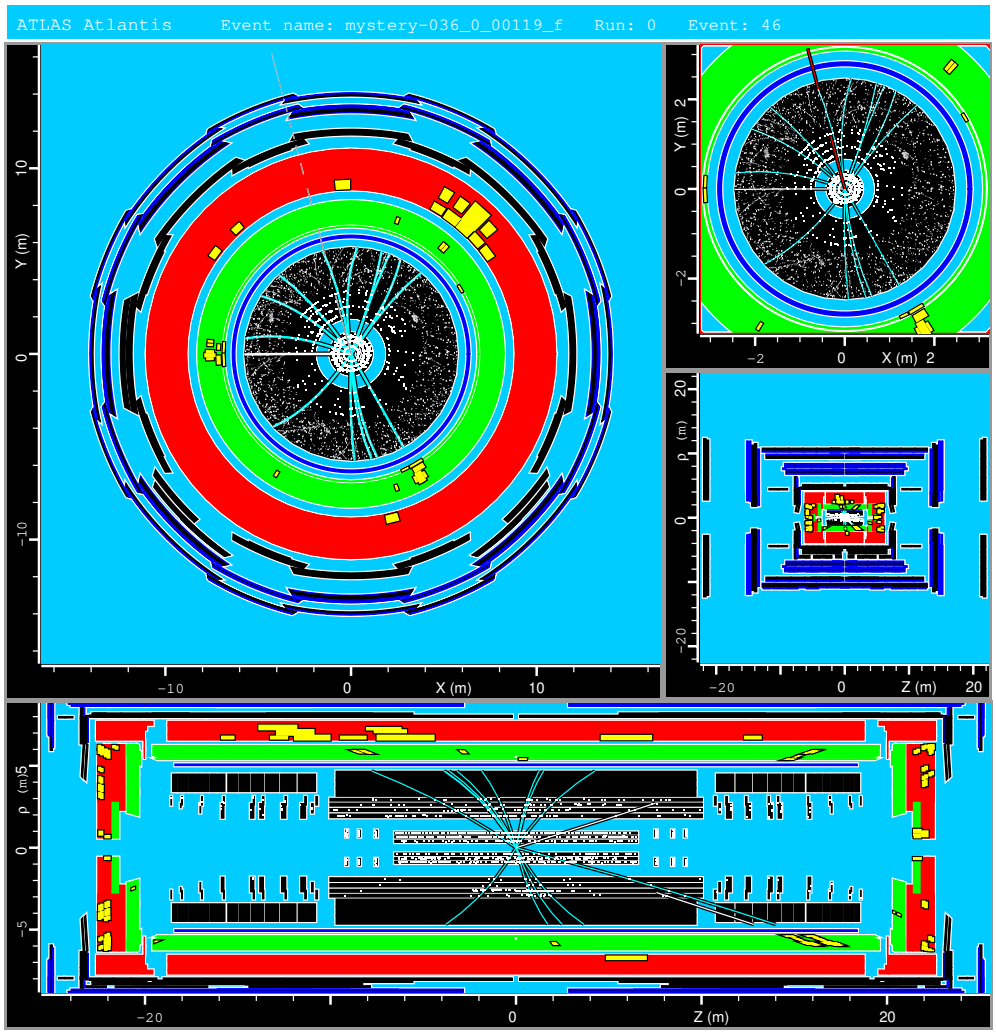

Figure 17: Event 119 with two high energetic electrons and jets

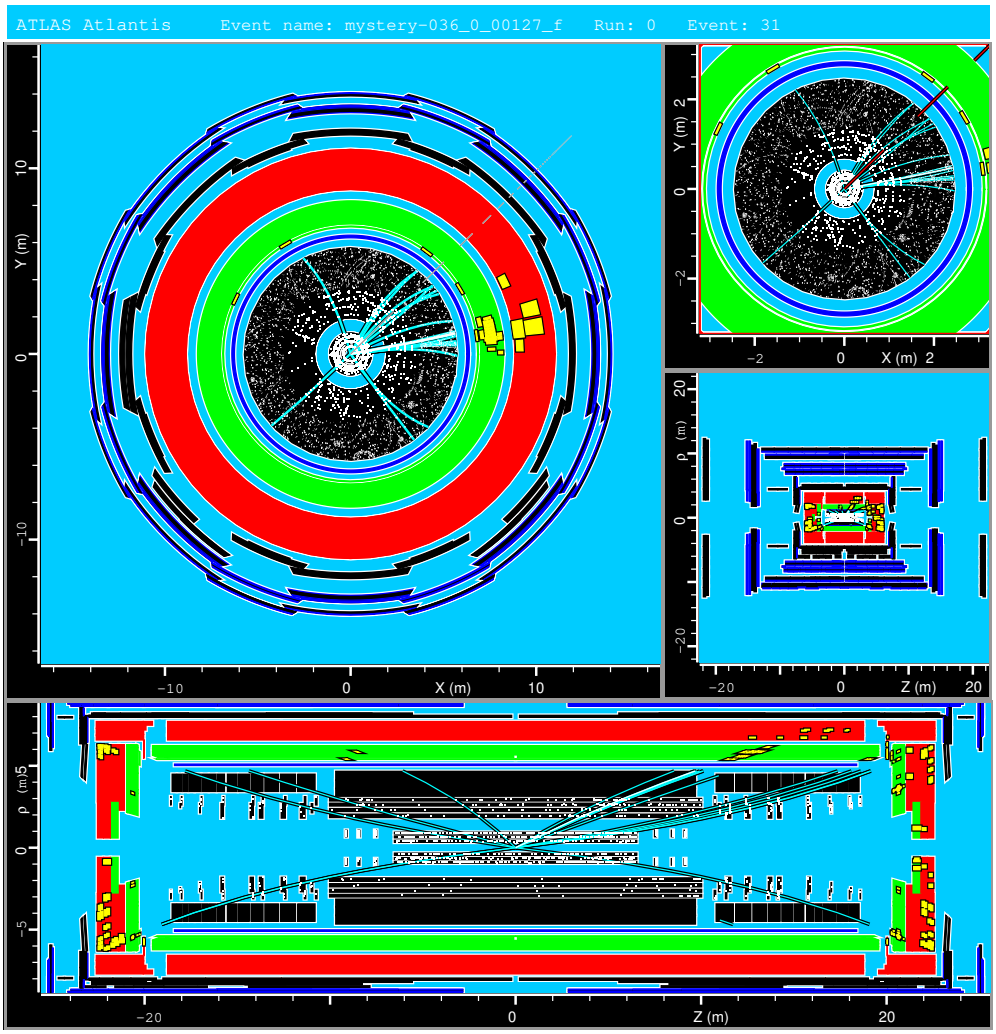

Figure 18: Event 127 with two high energetic electrons and jets

References

[1] CERN: CERN – European Organization for Nuclear Research. https://www.cern.ch. Accessed: 2026-03-02 (2025)

[2] Advanced Laboratory Course Physics 601: E214: The ATLAS Experiment (Properties of W Bosons and the Search for New Physics). University of Bonn, (2025). University of Bonn. Lab course manual, updated July 2025

[3] Group, P.D., Workman, R.L., *et al.*: Review of particle physics. Prog. Theor. Exp. Phys. **2024**(8), 083–01 (2024) https://doi.org/10.1093/ptep/ptae077